Lab 4

Greedy Best-First Search

Imagine a person standing at a place called S. The person wants to reach a destination called G. There are different roads that can be taken to reach the destination. The person does not know the exact best route, so they use a simple idea: always move towards the place that looks closest to the destination.

From S, there are two choices, One road goes from S to A, and another road goes from S to B. The road from S to A has a count of 2, while the road from S to B has a cost of 1. However, the person does not choose between A and B by looking at these road costs. Instead, they look at how close A and B appear to the final destination G.

A is estimated to be 2 units away from G, while B is estimated to be 3 units away from G. Since A appears to be closer to G, the person chooses A first. This is the main idea of Greedy Best-First Search: whenever there are different choices, it chooses the one that looks closest to the destination.

After reaching A, there is road from A to G. The cost of this road is 10. The person reaches G through the path S → A → G. The actual cost of this journey is 2 + 10 = 12.

At first, this might seem like a good choice because A looked closer to G. However, there was another route that the person did not choose first. From S, they could have gone to B. The road from S to B costs 1, and from B to G the cost is 4. Therefore, the total cost of S→B→G is only $1 + 4 = 5$.

This means that the route through B is much cheaper than the route through A. The route through A costs 12, while the route through B costs only 5. However, the person using the greedy method chooses A first because A looked closer to the destination.

This is an important point about this method. It makes a decision based on what looks best right now. It does not carefully compare the total cost of the entire journey before making its decision. Because of this, it can sometimes choose a route that looks promising at the beginning but turns out to be expensive later.

We can understand this with a simple example of travelling to a shopping mall. Imagine you are standing at a crossroads and want to reach the mall. You can see two roads. One road appears to take you very close to the mall, while the other road initially goes in a slightly different direction. You choose the first road because it looks closer to the mall. After travelling for sometime, you discover that this road has a very expensive toll. The other road would actually have been much cheaper. You made your decision based on what looked closest, without considering the complete journey.

The same thing happens here. From S, A appears to be closer to G because its estimated distance is 2, while B has an estimated distance of 3. Therefore, A is selected first. But the road from A to G is very expensive, costing 10. The route through B is much cheaper because it costs only 5 in total.

The numbers 4, 2, 3 and 0 are estimates of how far each place appears to be from the destination G. The destination G itself has an estimate of 0 because once we reach G, there is no distance left to travel. These estimates help the person decide which direction looks more promising.

It is important to understand that these numbers are only guesses or estimates. They do not necessarily tell us the real cost of reaching the destination. For example, A looks very close to G because its estimate is 2, but the actual road from A to G costs 10. B looks slightly farther away because its estimate is 3, but the actual route from B to G costs only 4.

The person therefore reaches G through S→A→G and spends 12. If the person had chosen B instead, the journey would have been S→B→G, costing only 5. So the greedy choice was not the cheapest choice.

Now we can see how the given program represents this idea. The places S, A, B and G are stored along with the roads connecting them. Each road has a cost. Another set of numbers gives the estimated distance of each place from G.

The search starts from S. The program looks at the estimated distances of the available choices. A has an estimated distance of 2, while B has an estimated distance of 3. Since 2 is smaller than 3, A is chosen first. The search then reaches G from A.

Only after reaching the destination does the program calculate how much the complete journey actually cost. The selected path is:

$S \rightarrow A \rightarrow G$

Final Comparison:

Greedy route: $S \rightarrow A \rightarrow G$
Actual cost: $2 + 10 = 12$

Cheaper route: $S \rightarrow B \rightarrow G$
Actual cost: $1 + 4 = 5$